But quantum physics doesn't let you calculate the momenta of A and B once they collide. Instead, the quantum wave function connects A and B in a strange way. Because of their collision, A and B share a single wave function, rather than having their own individual wave functions. But that shared wave function doesn't say what the particles' momenta are before a measurement is made. It simply ensures that, once A's momentum is measured, B's momentum will always be equal and opposite.

According to the Copenhagen interpretation, particles don't have definite properties until those properties are measured. So if A and B have definite momenta before they're measured, then the Copenhagen interpretation is wrong and quantum physics is an incomplete description of nature. But if A and B *don't* have definite momenta before they're measured, then the very act of measuring A's momentum must affect B instantly, in order to ensure that its momentum is equal and opposite to A's —even if A is in New York City and B is on the Moon. And that violates locality. In short, quantum physics is either incomplete or nonlocal. This forced choice was what had been "smothered" in the EPR paper, according to Einstein.

Einstein rejected any violation of locality, calling it "spooky action at a distance" in a letter to Max Born. He pointed out that there was no reason to assume any weird connection of this sort—the facts at hand could easily be explained by the incompleteness of quantum theory:

> When I consider the physical phenomena known to me, and especially those which are being so successfully encompassed by quantum mechanics, I still cannot find any fact anywhere which would make it appear likely that [locality] will have to be abandoned. I am therefore inclined to believe that the description of quantum mechanics in the sense of [the Copenhagen interpretation] has to be regarded as an incomplete and indirect description of reality, to be replaced at some later date by a more complete and direct one.

Meanwhile, the rest of the physics community was shocked by the EPR paper. "Now we have to start all over again, because Einstein proved that it does not work," wailed Dirac. In a fury, Pauli wrote a letter to Heisenberg, calling Einstein's behavior a "disaster" and asking Heisenberg to write a reply for publication. When Heisenberg heard that Bohr was working on his own reply, though, he shelved his draft and let the master himself respond